Where Cross-Question Structure in LLM Errors Comes From: Shared Seeds, Shared Prompts, and Answer-Letter Preference

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Abstract

Disagreement among a language model's samples, or among the models of an ensemble, is read as epistemic uncertainty: models that lack knowledge should miss the same groups of questions. We ask when such structure across the questions of a correctness matrix reflects shared knowledge gaps, and when it comes from the evaluation pipeline or from answer-letter preference; repeated samples of one model serve as controls. We count eigenvalues of the question correlation matrix above those produced by independent answers or, across models, by a single ability per model. Within one model, independent requests show no detectable direction, while a seed or exemplar set shared across one round of samples creates directions. Across 24 greedily decoded models, the first direction follows ability and the second is answer-letter preference: when the options are rotated it follows the displayed letter, not the question, and it rests mostly on one model. Counting a question correct only if it is answered correctly in all four rotations leaves no excess at the second eigenvalue, with low power against a second ability of typical strength; a marginal excess at the third and fourth comes mostly from two near-duplicate model pairs, partly from the letter-biased model. One letter-biased model can thus pose as a second dimension of knowledge; options should be rotated before an ensemble's error structure is read as epistemic.

1. Introduction

Self-consistency samples a language model K times on each question and reads disagreement among the samples as uncertainty about the answer (Wang et al., 2023; Manakul et al., 2023; Farquhar et al., 2024). Disagreement among the members of a deep ensemble (Lakshminarayanan et al., 2017) is read the same way, as epistemic uncertainty, which comes from missing knowledge, rather than aleatoric uncertainty, the noise that remains when the model is known (Kendall & Gal, 2017; Hüllermeier & Waegeman, 2021). A model that lacks some knowledge fails on the questions that need it, and errors of different models are related: models often choose the same wrong answer (Kim et al., 2025), and the scores of many models are summarized by a few latent factors (Burnell et al., 2023; Ruan et al., 2024; Maimon et al., 2026). We call the tendency of questions to be answered right or wrong together *cross-question structure*, and ask when it is evidence of shared knowledge gaps and when it is an artifact of the pipeline (a shared seed, shared exemplars) or of answer-letter preference.

The kth samples of all questions form *pass* k; like one ensemble member, a pass answers every question once. Self-consistency samples each prompt by a separate request with its own random numbers, so given the model, correctness on two questions of a pass is independent by construction. A pipeline can still couple them: one seed in a batched call's sampling parameters gives every prompt the same random numbers, and the standard 5-shot MMLU prompt places one fixed set of exemplars before the questions of a subject (Hendrycks et al., 2021). Within one model, a test for such structure is therefore a control: independent requests should show no structure, and a shared seed or exemplar set should. Different models share no sampling noise, but they differ in ability and in habits unrelated to knowledge: a model that prefers one answer letter is right more often on the questions whose correct option carries that letter (Zheng et al., 2024; Pezeshkpour & Hruschka, 2024).

We test a correctness matrix, passes or models by questions, for eigenvalues of its question correlation matrix above what independent answers produce (Marčenko & Pastur, 1967; Johnstone, 2001) or, across models, one ability per model (Rasch, 1960; Lord & Novick, 1968). Each such eigenvalue marks a shared *direction*: a pattern of questions along which the rows vary together.

Contribution and related work. The tools are classical (App. A), and spectral dimensionality measures compared

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with a Marchenko–Pastur null, together with parallel analysis, have been applied to item-level benchmark matrices, where binary spectra are known to overstate dimensionality (Sha & Zhao, 2026). Letter preference and order sensitivity are known for single models (Zheng et al., 2024; Pezeshkpour & Hruschka, 2024), as is the influence of few-shot exemplars on the answer distribution (Zhao et al., 2021; Min et al., 2022). Ding (2026) shows with an option-shuffle control that part of a model's self-agreement is positional. In the option-level response model of Fei et al. (2026), a per-model positional preference barely helps to predict held-out correctness (prediction accuracy 0.840 with it, 0.838 without); in a binary correctness matrix, which records no letter, we find that removing one strongly biased model brings the second eigenvalue down to within what one ability produces. Our contribution is not the method but its controls and attribution: protocols that calibrate the test, an option-rotation design, scored by the CircularEval rule of Liu et al. (2024), that attributes the second direction to letter preference, and validity-checked one-ability nulls with measured power and, for the main across-model test, size.

2. A spectral test for cross-question structure

Entry c_{ki} of the correctness matrix $\mathbf{C}$ is 1 when row k answers question i correctly and 0 otherwise. Row k is pass k within one model and model m across models. A question that every row gets right, or none does, carries no correlation, so we keep the questions that vary: within a model, the N_b *borderline* questions with pass rate $\varepsilon < f_i < 1 - \varepsilon$, $\varepsilon = 0.05$; across models, the *split* questions, which some models answer correctly and others do not. Let $\mathbf{R}$ be the Pearson correlation matrix of the kept columns over the rows. If the questions are independent, the eigenvalues of $\mathbf{R}$ spread around 1 and, as rows and columns grow with their ratio γ (kept columns over rows) fixed, follow the Marchenko–Pastur (MP) law, whose upper edge is $\lambda_+ = (1+\sqrt{\gamma})^2$ (Marčenko & Pastur, 1967). An eigenvector gives the question loadings of its direction; a row's *score* on it is the projection of its standardized row.

Within one model. For Gaussian data, the largest eigenvalue $\lambda_{\max}$ fluctuates near λ_+ on the Tracy–Widom scale σ_{TW}, the size of its typical fluctuation (Johnstone, 2001), and $z = (\lambda_{\max} - \lambda_+)/\sigma_{\mathrm{TW}}$ measures its distance above the edge. Because no theorem we know gives this limit for sample-centered binary entries (App. A), we compare each z with a Monte Carlo null of 1000 matrices of independent Bernoulli entries at the observed pass rates, with the filter re-applied (App. C). The MP count is the number of eigenvalues above λ_+; the buffered count, above $1.05\,\lambda_+$, leaves a margin for the fluctuation of $\lambda_{\max}$ and is positive in 0.65% and 0.35% of simulations without shared structure at the pass rates of Qwen2.5-7B and Llama-3-8B on MMLU (App. D).

Across models. Here λ_+ is not the reference: a one-ability model with a logistic response can produce a second eigenvalue above λ_+ by itself (Sec. 3.3). We therefore compare each eigenvalue λ_j with the jth eigenvalue of 1000 matrices simulated from a one-ability model fitted to the data. In the two-parameter logistic (2PL) model (Lord & Novick, 1968), model m answers question i correctly with probability $\sigma(a_i(\theta_m - b_i))$, with one ability θ_m per model and a difficulty b_i and slope a_i per question; Rasch sets every slope equal (Rasch, 1960), and the 3PL adds a guessing floor of 0.25. In the *letter-aware* null, a model knows the answer with the 2PL probability and otherwise guesses letter L with its own probability $\pi_m(L)$; the *uniform know-or-guess* null guesses uniformly. A null rejects rank j when λ_j exceeds the 99th percentile of its jth eigenvalue; ranks $2, 3, \ldots$ rejected in a row are *directions beyond ability*. A null counts as evidence only if it is *valid*, that is, it reproduces the ability eigenvalue: the observed λ_1 lies between its 1st and 99th percentiles. An excess can also spread over several ranks. The *family-wise test* therefore takes the largest percentile of the observed λ_2, λ_3 and λ_4 among the null draws, and compares it with the same maximum in each of 2000 new null draws, giving its p-value p_T (App. F). We fixed this statistic, its ranks, the sibling groups of Sec. 3.3 and the validity rule before the test's run, after an audit of an earlier draft of this analysis had computed a similar statistic on the same data. The first analyses used an earlier validity convention; the pair and single-model follow-ups are post hoc.

Runs. Within one model (App. B), the *independent* protocol gives every (question, pass) pair its own seed, as self-consistency does; in the *shared-seed* protocol all questions of a pass use one seed; in the *shared-prompt* protocol each pass places its own five MMLU development questions, with their answer letters, as exemplars before every question. Twelve independent-draw runs cover five instruction-tuned models on MMLU (Hendrycks et al., 2021) and three on HellaSwag (Zellers et al., 2019) at temperature 1, and Qwen2.5-7B on GSM8K (Cobbe et al., 2021) at two output lengths and on MMLU at temperatures 0.5 and 1.5 (Table A1); three *shared-component* runs, one shared-seed and two shared-prompt, use MMLU with $K = 100$. Across models, 24 models (23 instruction-tuned, one base) of 0.5B to 22B parameters from ten families answer the 500 MMLU questions by greedy decoding, in four *rotations*: rotation r shows original option j at letter position $(j + r) \bmod 4$; rotation 0 is the original.

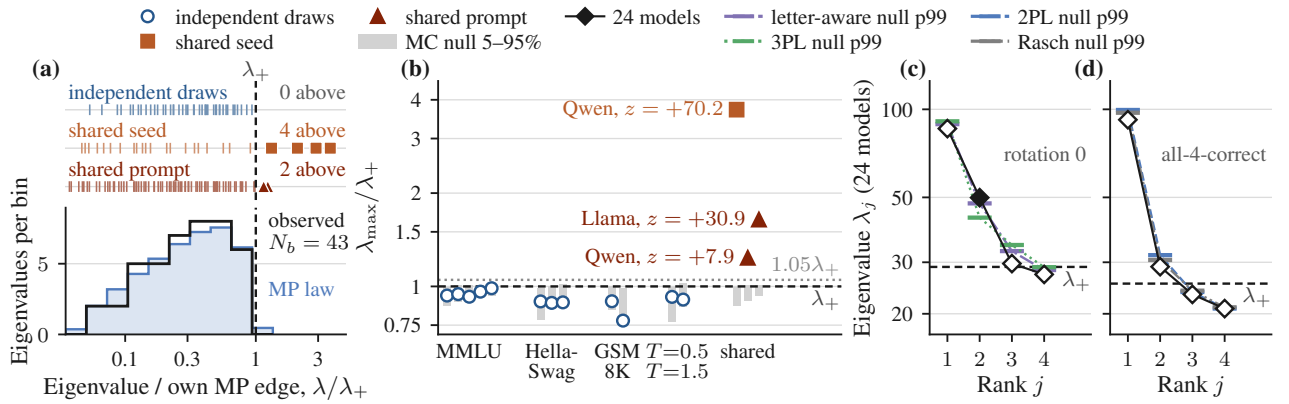


Figure 1. **(a)** Eigenvalues of $\mathbf{R}$ for three runs of Qwen2.5-7B on MMLU ($K = 100$): independent draws, shared seed and shared prompt, divided by each matrix's own MP edge λ_+ (dashed); markers: above the edge. Lower: histogram of the independent-draw run's eigenvalues ($N_b = 43$) and the MP law. **(b)** $\lambda_{\max}/\lambda_+$ for every within-model run (Table A1); grey bars: 5–95% range of each run's Monte Carlo null; dotted: $1.05\,\lambda_+$. **(c,d)** The four largest eigenvalues of the 24-model matrix (c) in rotation 0 and (d) counting a question correct only if correct in all four rotations, against the rank-matched 99th percentiles of two one-ability nulls each: (c) the letter-aware null and the 3PL; (d) the 2PL and Rasch. Only valid nulls are shown. Filled marker: above both nulls; dashed: λ_+. The figure and the per-rank results of Sec. 3.2 use 1000 draws per null; the family-wise p_T and percentiles of Sec. 3.3 on the full data, 2000 new draws; App. F gives the draw counts of all other results. In (d), $\lambda_4 = 20.83$ equals the 2PL rank-4 99th percentile (20.85) within Monte Carlo error.

3. Results

3.1. Controls within one model

In all 12 independent-draw runs the largest eigenvalue lies below the MP edge (Fig. 1a,b), and every z lies between the 10th and the 84th percentile of its Monte Carlo null; the counts stay zero at six borderline thresholds (App. C). The test finds no hidden coupling in our pipeline, within its power: in simulations where each answer is decided by a normal variable and these variables correlate at ρ across a pass's borderline questions (App. D), at $\rho = 0.05$ the MP count detects the coupling in every matrix at Llama-3-8B's $N_b = 173$ and in 47% at Qwen2.5-7B's $N_b = 43$, and a nine-question cluster at $\rho = 0.2$ in at most 23%.

In the shared-component runs, z exceeds every draw of its null and two to four eigenvalues exceed $1.05\,\lambda_+$; the directions follow the shared input (App. E). A shared seed gives one answer letter the same random boost in every question of a pass, so gold letter explains 73% of the variance of the top eigenvector's loadings ($\eta^2 = 0.73$, permutation $p = 0.0002$). Llama-3-8B copies the shared exemplars' answer letters ($\eta^2 = 0.53$, $p = 0.0002$). Qwen2.5-7B's shared-prompt direction does not group by gold letter ($p = 0.094$), but two passes whose exemplars share a question answer more alike than other pairs ($p = 0.0005$).

3.2. Across models: a second direction, from letter preference

In rotation 0 the four largest eigenvalues of the 24-model matrix (460 split questions) are 85.9, 49.9, 29.7 and 27.3, against $\lambda_+ = 28.92$. The models' scores on eigenvector 1 correlate with accuracy at $r = 0.94$: it is the ability direction. Four nulls are valid: the letter-aware, 3PL and uniform know-or-guess nulls (λ_1 at their 86.7th, 76.4th and 98.7th percentiles) and the soft-preference null (below); the 2PL and Rasch put λ_1 above all their draws. Each of the first three rejects rank 2 and no other (Fig. 1c), and the family-wise test rejects all three ($p_T = 0.002$).

Eigenvector 2 is answer-letter preference. Its question loadings group by gold letter ($\eta^2 = 0.48$, $p = 0.0002$), and the models' scores on it correlate with their fraction of answers that are D at $r = -0.90$ (Fig. A3a) and with their fraction that are A at $r = 0.76$. Within one rotation a question's gold letter is tied to its content, so this grouping cannot separate the two; rotating the options can. In rotations 1–3 we correlate each question's loading on eigenvector 2 (matched by the models' scores) with its own rotation-0 loading and with the rotation-0 mean loading of questions whose gold letter is its new one. A direction of question content would follow the own loading; this one follows the shown letter ($r = 0.60$ to 0.64, against -0.06 to 0.16; Fig. A3b).

The direction rests mostly on one model, Llama-3.2-1B, whose estimated letter preference when it guesses puts probability 0.90 on A. Removing it lowers λ_2 from 49.9 to 35.0, below the 99th percentiles of the 3PL and letter-aware nulls

refitted to the other 23 models (40.1 and 38.1), so the per-rank rule finds no direction beyond ability; removing any other single model leaves λ_2 between 47.1 and 52.4. A post hoc family-wise test still rejects both nulls ($p_T = 0.0035$ and 0.015): a weaker excess remains at ranks 3–4, where the letter grouping moves (eigenvector 3, $\eta^2 = 0.21$).

The letter-aware null lets the preference act only on guesses and under-reproduces λ_2. In a *soft-preference* (SP) model the preference also tilts choices a model partly knows, and ability is the only link between questions; the letter-aware null rejects rank 2 in 60% of datasets simulated from it. The SP null is also valid (λ_1 at its 85.7th percentile); it rejects rank 2 (calibrated $p = 0.010$) but not family-wise ($p_T = 0.079$), and its fit fails a predictive check (App. F). Neither model thus reproduces all of λ_2, but the rotation test leaves little room for a part that follows question content: we read the remainder as letter preference that neither model captures fully, not excluding a small content part.

3.3. All-4-correct scoring: a marginal excess, mostly from two model pairs

To remove letter preference, we count a question correct for a model only when the model answers it correctly in all four rotations, the CircularEval rule (Liu et al., 2024). The gold option appears once at each letter over the rotations, so a model that only guesses by letter is correct in all four with probability $\prod_L \pi_m(L) \leq 1/256$, the same for every question. In this *all-4-correct* matrix (391 split columns, $\lambda_+ = 25.36$) the eigenvalues are 92.1, 29.0, 23.3 and 20.8 (Fig. 1d). The valid nulls are the 2PL and Rasch (λ_1 at their 31.5th and 54.2nd percentiles); the other nulls miss λ_1. The 99th percentile of λ_2 is 31.9 under the 2PL and 30.2 under Rasch, above λ_+, so one ability alone can lift a second eigenvalue above the edge; the observed λ_2 lies below both: no second direction separates at rank 2.

The excess lies lower: λ_3 and λ_4 sit at the 98.5th and 99.1st percentiles of the 2PL and the 97.0th and 98.6th of Rasch, and the family-wise test rejects both nulls ($p_T = 0.026$ and 0.041). Under one-ability truth the test rejects in up to 8.7% of datasets at the 5% level; calibrated against them, the observed p_T are 0.017 to 0.057, a marginal excess (App. F).

It comes mostly from two near-duplicate model pairs. Keeping one model of each of four sibling groups (versions or fine-tunes of one base model at one size), in all 24 combinations, removes the rejection, while removing as many other models at random does not (App. F). A post hoc follow-up narrows this to two pairs: removing one model each of Phi-3-mini/Phi-3.5-mini and Meta-Llama-3-8B/Llama-3.1-8B gives p_T 0.22 to 0.69 (2PL) and 0.32 to 0.38 (Rasch), above all 190 removals of two other models (at most 0.099 and 0.133). Removing the near-duplicates Mistral-7B v0.2 and v0.3 keeps the rejection ($p_T = 0.019$ under both). The pair excess may reflect knowledge gaps shared through common training data, or another shared artifact such as contamination; we did not test which. Without Llama-3.2-1B alone the family-wise test no longer rejects (p_T 0.06 and 0.089), so part of the excess depends on that model. No factor shared across the ensemble is detected, at low power.

Three limits qualify this result (App. F). First, all-4-correct removes a preference only if it acts when a model guesses; the SP null, which lets it act on every choice, is not valid on all-4-correct. Second, scores that do not bound guessing give no reliable verdict: on "correct in at least two (or three) of four", the 2PL test rejects one ability, but also in 98.5% (58%, indicative only, from an invalid fit) of datasets simulated from the SP fit, which has no second factor; not every valid null rejects. We chose all-4-correct for its guessing bound, not for its verdict. Third, power is low: a second ability planted on the 180 STEM questions with about a typical fitted slope is detected in 6% (2PL) and 8% (Rasch) of datasets by the per-rank rule and in 34 and 21% by the family-wise test (indicative rates: invalid SP fit).

4. What is not shown

Letter preference and the two pairs are the largest structure beyond ability we find, not necessarily the only one. We test correctness only, not log-probabilities, answer distributions, semantic clusters or reasoning traces. The test cannot exclude small clusters of questions within a model, nor a weaker second factor across models (Sec. 3.3); power for structure carried by a few models was not simulated. Architecture, family, scale, data, instruction tuning and decoding vary together in the ensemble; one architecture trained with several seeds, or a dropout mask shared by a pass (Gal & Ghahramani, 2016), would isolate a training seed or a shared parameter perturbation. The within-model runs cover only instruction-tuned models of 1.7B to 8B parameters, and the ensemble 500 MMLU questions. Per-rank thresholds are uncorrected over the nulls and scores of Table A2.

5. Discussion

One strongly letter-biased model can pose as a second dimension of knowledge in a correctness matrix. Item-level factor and item response analyses (Kipnis et al., 2025; Sha & Zhao, 2026; Yao et al., 2026) and agreement on the same wrong answer (Kim et al., 2025) may contain the same confound; we have not tested them. Before reading an ensemble's errors as epistemic, as in cross-model uncertainty scores (Hamidieh et al., 2026), one should rotate the options (four answers per question), score a question correct only in all rotations, and test the remaining spectrum family-wise against valid one-ability nulls, with near-duplicates in mind.

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A. Background

This appendix describes the sampling-based uncertainty methods whose samples' correctness the paper tests, and states the classical results behind the tests: random-matrix limits for independent variables and item response models.

Sampling-based uncertainty. Sampling-based methods for language models (Wang et al., 2023; Manakul et al., 2023; Farquhar et al., 2024) estimate uncertainty from the spread of one model's samples, and Cecere et al. (2025) study how the sampling temperature affects these estimates; a model can also be asked to evaluate whether its own answers are true (Kadavath et al., 2022). In a correctness matrix, the spread of one model's samples on question i is the Bernoulli variance $p_i(1-p_i)$, set by that question alone. The paper asks whether this spread also has the structure of an ensemble's disagreement, in which members are right or wrong on groups of questions together. Gu et al. (2026) show that language models do not reliably sample from a target distribution when asked to, a different property of sampled output from the one tested here.

Random-matrix tests for the number of factors. A factor shared by many variables lifts an eigenvalue of their sample correlation matrix beyond the Marchenko–Pastur edge λ_+ of Section 2: n variables with pairwise correlation κ have a population eigenvalue of $1+(n-1)\kappa$. This eigenvalue grows with n, so an eigenvalue test can detect a correlation smaller than the sampling noise of one pairwise correlation, about $1/\sqrt{K}$. Comparing sample eigenvalues with the MP edge is how Laloux et al. (1999) and Plerou et al. (1999) separate market factors from noise in stock-return correlations. The MP law holds for the sample covariance matrices of independent entries with finite variance (Marčenko & Pastur, 1967; Bai & Silverstein, 2010) and, for i.i.d. entries, for their sample correlation matrices (Jiang, 2004). Johnstone (2001) showed for Gaussian data that the largest eigenvalue, minus the edge and divided by $\sigma_{\mathrm{TW}} = (1+\sqrt{\gamma})(1+1/\sqrt{\gamma})^{1/3}K^{-2/3}$, converges to the Tracy–Widom distribution. For sample correlation matrices, Bao et al. (2012) proved this limit for entries with a symmetric distribution and no centering, with fewer variables than observations, and with centering only for Gaussian entries; Pillai & Yin (2012) removed the symmetry condition and allowed more variables than observations, still without centering. Our $\mathbf{R}$ is centered at the sample means of Bernoulli columns with unequal pass rates; no theorem we know covers sample-mean centering of non-Gaussian entries, so the Monte Carlo null, not the limit, calibrates z.

Item response models. Item response models come from educational testing, where each test-taker has an ability and each question a difficulty; here the test-takers are language

models. In the Rasch model (Rasch, 1960) model m answers question i correctly with probability $\sigma(\theta_m - b_i)$, with σ the logistic function: the probability rises with ability θ_m, falls with difficulty b_i, and is $1/2$ when the two are equal. The 2PL model (Lord & Novick, 1968) adds a slope a_i, giving $\sigma(a_i(\theta_m - b_i))$; a larger slope makes the probability rise more sharply with ability. The 3PL form adds a guessing floor, here $0.25 + 0.75\,\sigma(a_i(\theta_m - b_i))$ for four options. All are one-factor models: ability is the only source of dependence between questions. Such a model can still place more than one eigenvalue of a 24-row correlation matrix above λ_+ (Section 2), and the all-4-correct eigenvector 2 follows question difficulty (App. F).

B. Data collection

Every matrix in the paper comes from the protocol below; the last paragraphs check the scoring of unparseable replies and the truncation on GSM8K, and record how the data were collected.

Questions. MMLU: 500 test questions, ten drawn at random from each of the first 50 subjects in alphabetical order. HellaSwag: 200 validation items, a random sample stratified by activity label. GSM8K: the first 100 test problems. Each question is matched to its exact dataset row. The shared-prompt exemplars come from the 285 MMLU development questions: pass k uses five drawn at random from all subjects, independently of the other passes, so two passes can share an exemplar. The set is the same for every question of the pass and in both shared-prompt runs.

Prompts and decoding. Every model receives one user message rendered with its own chat template; Phi-2, which has none, receives the raw text. Prompts are tokenized with each model's own tokenizer and checked against an independent reference tokenization. In the shared-prompt protocol, the five exemplars, each followed by its answer letter, sit in this message between the instruction and the question. Generation uses vLLM 0.23.0 in bfloat16 with temperature T, top-$p = 1$, top-k disabled, min-$p = 0$, repetition penalty 1.0, no presence or frequency penalty, and at most 8 new tokens for MMLU and HellaSwag and 256 (or 1024) for GSM8K. Greedy decoding is $T = 0$. In the independent and shared-prompt protocols, every sample (k, i) has its own seed, distinct from every other seed of the run; in the shared-seed protocol, all questions of pass k share one seed, and the seed changes from pass to pass. vLLM gives each seeded request its own random-number generator, so a request's sampling noise depends on its seed, not on its position in a batch, and requests with different seeds do not share noise (App. E).

Rotations and replicates. The 24 models answer every question in all four rotations, with deterministic kernel settings. GPU kernels can still differ between engine starts, so the ensemble was collected twice, with separate engine starts (replicates A and B). The paper reports replicate A.

Scoring. A multiple-choice reply is read as letter X when it begins with X, after optional "Answer:", parenthesis or bold markers, followed by punctuation, a line end, or words that do not negate or hedge the letter; or when its first line says "the answer is X" or a similar phrase. A reply that names several letters, negates its letter, or cannot be read this way is unparseable and scored wrong. The rule is fixed by unit and property-based tests. The GSM8K prompt asks for the final answer after "####"; a reply is correct when the number after its last "####", or else its last number, equals the gold answer.

Unparseable replies. In the sampled runs at most 1.65% of replies are unparseable (Llama-3-8B on HellaSwag), and the shared-seed run and Qwen2.5-7B's shared-prompt run have none. In the ensemble, Mistral-7B v0.2 has 12.4% and every other model at most 9.6%. We either drop every question with an unparseable reply or replace each unparseable reply by the mean correctness of the question's parseable replies. Either way, every count of the independent-draw runs stays at zero, and the MP and buffered counts of Llama-3-8B with a shared prompt stay at two. Across models, we rerun the valid nulls on both variants (Table A4). On all-4-correct, the family-wise rejection survives imputation and dropping the 95 questions whose all-4-correct score depends on an unparseable reply, but not dropping all 217 questions with any unparseable reply, which keeps 236 of 391 columns and has less power. In rotation 0, imputation leaves rank 2 above the letter-aware and 3PL 99th percentiles; dropping the 112 affected questions moves it to the 98.3rd percentile of the letter-aware null, while the 3PL still rejects it and the family-wise test rejects under both ($p = 0.013$ and 0.004).

Truncation on GSM8K. At 256 new tokens, 55% of Qwen2.5-7B's chain-of-thought replies reach the limit. Most of them stop before the final answer, and the scorer then reads the last number of an unfinished calculation: 13.6% of truncated replies are correct, against 96.7% of complete ones. Accuracy is 51.0% at 256 tokens and 90.3% at 1024 tokens. The run still tests independence: truncation changes which answers are correct, not whether two questions of a pass share a cause.

Collection history. The data were collected anew after two rounds of checks. The first found a scorer that accepted a reply beginning with the gold letter as a word ("Based on ..." as B) and a question loader that matched questions by a text prefix shared by some MMLU questions; the second

Table A1. The test on every within-model run ($\varepsilon = 0.05$). λ_+: MP edge; TW z: Tracy–Widom statistic. Counts: MP / buffered. GSM8K runs use $T = 1$.

Run	N_b	λ_+	$\lambda_{\max}$	TW z	Counts
Independent, MMLU ($K = 100, T = 1$)					
Qwen2.5-7B	43	2.74	2.56	-1.72	0/0
Mistral-7B-v0.3	83	3.65	3.45	-1.79	0/0
Llama-3-8B	173	5.36	4.97	-3.05	0/0
SmolLM2-1.7B	406	9.09	8.74	-2.17	0/0
Phi-3-mini	271	7.00	6.91	-0.65	0/0
Independent, HellaSwag ($K = 50, T = 1$)					
Qwen2.5-7B	16	2.45	2.19	-1.60	0/0
Mistral-7B-v0.3	65	4.58	4.05	-2.71	0/0
Llama-3-8B	86	5.34	4.75	-2.90	0/0
Independent, Qwen2.5-7B ($K = 30$)					
GSM8K, 256 tok.	62	5.94	5.32	-2.06	0/0
GSM8K, 1024 tok.	19	3.22	2.50	-2.97	0/0
MMLU, $T = 0.5$	25	3.66	3.39	-1.08	0/0
MMLU, $T = 1.5$	85	7.20	6.53	-2.07	0/0
Shared component, MMLU ($K = 100, T = 1$)					
Qwen, seed	41	2.69	10.00	$+70.2$	4/4
Qwen, prompt	93	3.86	4.77	$+7.9$	2/2
Llama, prompt	264	6.89	11.31	$+30.9$	2/2

found that three tokenizers (DeepSeek-LLM-7B, Yi-1.5-6B, OLMo-2-7B) were loaded incorrectly, which changed those models' prompts in the ensemble. Each defect was fixed and covered by a test. The sampled runs were collected once with the fixed loader and scorer and re-scored with the final parser; the ensemble was collected again for all 24 models, in four rotations and two replicates.

C. Test statistics and checks

This appendix lists the within-model runs, states the independence that the within-model test assumes and what it implies for the spectrum, defines the Horn count, checks the independence on the data, and varies the borderline threshold.

Assumption C.1 (Independent draws). Given the model and the prompts, the entries c_{ki} are Bernoulli variables with $\Pr(c_{ki} = 1) = p_i$, independent across questions i and passes k.

Under Assumption C.1 the population correlation between two borderline questions is zero, so the population correlation matrix of the borderline columns is the identity. The filter keeps or drops each question on the basis of its own column alone, so the kept columns stay independent of each other. If every p_i stays a fixed distance from ε and $1-\varepsilon$, the kept set also equals the set of questions with $\varepsilon < p_i < 1-\varepsilon$ with probability tending to 1 as K and N grow in proportion, because the probability that f_i falls on the wrong side of a threshold decreases exponentially in K. Centering column i at p_i and dividing it by $\sqrt{p_i(1-p_i)}$ gives independent, bounded, mean-zero, unit-variance entries, but $\mathbf{R}$ centers at the sample means, and the columns are not identically distributed, because their pass rates differ (App. A). At $K \le 100$ the distribution of z is shifted from its limit: every z of the 12 independent-draw runs is negative, and their Monte Carlo nulls of z have means between -1.98 and -1.47. We therefore calibrate each run's z against its Monte Carlo null. A z above every draw of that null rejects Assumption C.1, as in the three shared-component runs, but does not say whether questions within a pass or the passes themselves are dependent.

Horn count. The Horn count (Horn, 1965; Buja & Eyuboglu, 1992) is the number of leading eigenvalues that each exceed the 95th percentile of the eigenvalue of the same rank in 200 surrogate matrices, made by shuffling every borderline column across passes, which keeps the pass rates and removes any correlation between questions. The count stops at the first rank whose eigenvalue does not exceed its percentile. It is 0 in all 12 independent-draw runs, and 5, 10 and 5 in the shared-seed run and the shared-prompt runs of Qwen2.5-7B and Llama-3-8B. Because these thresholds fall with rank while λ_+ stays fixed, the Horn count can count eigenvalues below λ_+ and exceed the MP count, as for Qwen2.5-7B with a shared prompt (10 against 2).

Checks of independence. Three further checks on the independent-draw runs agree with Assumption C.1. A component that makes the questions of a pass right or wrong together would make the number of borderline questions a pass answers correctly vary more than independent draws at the pass rates predict. The ratio of the observed to the predicted variance of this number lies between 0.61 and 1.09 in the 12 runs, and no run is significantly overdispersed against a column-shuffle null (smallest $p = 0.27$); the two lowest ratios have one-sided underdispersion $p = 0.039$ and 0.045, neither significant after a Holm correction over the 48 tests (12 runs $\times$ 4 statistics). This check cannot detect a coupling that cancels in the number correct, such as that of the shared seed (App. E). Second, in the $T = 1$ MMLU runs of Qwen2.5-7B and Llama-3-8B, borderline questions of the same subject are no more correlated than other pairs (mean same-subject correlation -0.018, $p = 0.80$, and 0.002, $p = 0.37$). Third, a pairwise test that compares the largest Cramér's V and mutual information over all pairs of borderline questions with their permutation null finds no coupled pair, in correctness in all five MMLU runs at $T = 1$, and in the answer letters of Qwen2.5-7B (largest V 0.41 against a null 99th percentile of 0.72). The samples still estimate each p_i: Qwen2.5-7B's pass rates from passes 1–50 and 51–100 correlate at $r = 0.997$ over the 500 questions and 0.940 over the 43 borderline ones.

Borderline threshold. The threshold ε decides which questions enter the test, and with them N_b, λ_+ and the

power. We repeat the test at $\varepsilon = 0.01$, 0.02, 0.05, 0.10, 0.15 and 0.20, with the filter re-applied at each threshold to the data and to every matrix of the run's Monte Carlo null (1000 matrices). Over all 12 independent-draw runs and six thresholds, no eigenvalue exceeds its edge, and every observed z lies below the 95th percentile of its null. The three shared-component runs keep at least two eigenvalues above their edge at every threshold. The six tests of a run are nested, not independent: raising ε removes the questions whose pass rates lie closest to 0 or 1. At $\varepsilon = 0.20$ the three smallest sets have 6, 7 and 10 questions (GSM8K at 1024 tokens, HellaSwag Qwen2.5-7B, MMLU at $T = 0.5$).

D. Power of the within-model test

This appendix measures which shared structure the null result of Section 3.1 can exclude. We simulate correctness matrices at the pass rates of two independent-draw MMLU runs, Qwen2.5-7B ($N_b = 43$) and Llama-3-8B ($N_b = 173$), with a planted factor that makes some questions of a pass right or wrong together, and measure how often each count detects it. The factor loads on all of a run's borderline questions (diffuse), or on 30 or 9 of them chosen at random in each simulated matrix. Each entry comes from a hidden standard normal score: $c_{ki} = 1$ when the score falls below $\Phi^{-1}(f_i)$, with Φ the standard normal distribution function, so question i is correct with probability f_i. For a loaded question the score is $\sqrt{\rho}\,u_k + \sqrt{1-\rho}\,e_{ki}$, where the factor u_k is one standard normal draw shared by the loaded questions of pass k and e_{ki} is independent standard normal noise; for the other questions the score is e_{ki}. The score keeps unit variance, so each question keeps its pass rate, and the scores of two loaded questions correlate at ρ. The correlation of their 0/1 entries is smaller: averaged over pairs of borderline questions, it is about half of ρ at these pass rates.

Each simulated matrix has all 500 questions and $K = 100$ passes and goes through the test of Section 2, with the borderline filter re-applied. A count detects the factor when it is at least one; the detection rate is the power for $\rho > 0$ and the false-positive rate for $\rho = 0$. Each scenario has 500 simulated matrices, and $\rho = 0$ has 2000 (500 for the Horn count, which uses 100 surrogates per matrix).

Power for a diffuse factor is higher at Llama-3-8B's 173 borderline questions, because the factor then loads four times as many questions; power for a fixed-size cluster is lower there, because the cluster is a smaller share of the questions (Fig. A1). At $\rho = 0.05$ the MP count detects the diffuse factor in 47% of Qwen2.5-7B simulations and in 100% of Llama-3-8B simulations. A 9-question cluster at $\rho = 0.2$ is detected in 23% (MP) and 26% (Horn) of Qwen2.5-7B simulations and in 11% (MP) and 8% (Horn) of Llama-3-8B simulations. At $\rho = 0$ the false-positive rates are 4.6% and 8.3% (MP), 0.65% and 0.35% (buffered), and 6.4% and 5.4% (Horn). Runs with other numbers of borderline questions were not simulated; we expect power for a diffuse factor to fall, and for a fixed cluster to rise, as N_b shrinks.

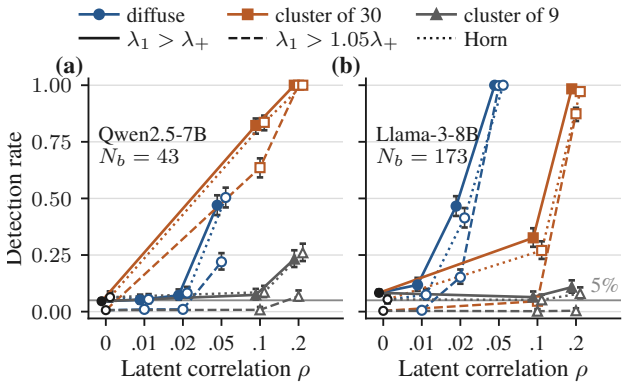


Figure A1. Detection rate (fraction of simulated matrices with a count of at least one) against the latent correlation ρ of a planted factor, at the pass rates of (a) Qwen2.5-7B ($N_b = 43$) and (b) Llama-3-8B ($N_b = 173$) on MMLU. The ρ values are placed at evenly spaced positions, not to scale. Color: where the factor loads; black markers at $\rho = 0$: no factor. Line style: count (solid, filled markers: MP; dashed, open: buffered; dotted, open: Horn). Whiskers: Wilson 95% intervals. Grey line: a detection rate of 5%.

E. What the shared component couples

This appendix explains how each shared component couples the questions of a pass, and measures what the resulting directions follow.

Shared noise under a shared seed. At each generation step, the random-number generator of a seeded request draws one exponential random number, the noise, for every token of the vocabulary, and the request emits the token whose probability at temperature T, divided by its noise, is largest. This is Gumbel-max sampling, since the negative logarithm of an exponential random number is a Gumbel random number: each answer, taken alone, follows the model's distribution for its question. Two requests with the same seed draw the same noise vector at every step, whatever their prompts; coupled generation and counterfactual sampling use this sharing on purpose (Corvelo Benz et al., 2026; Ravfogel et al., 2025). If a shared-seed pass draws a small noise value for the token "B", the sampler divides the probability of "B" by that small value in every question of the pass whose letter is generated at that step, and so favors "B" in all of them at once.

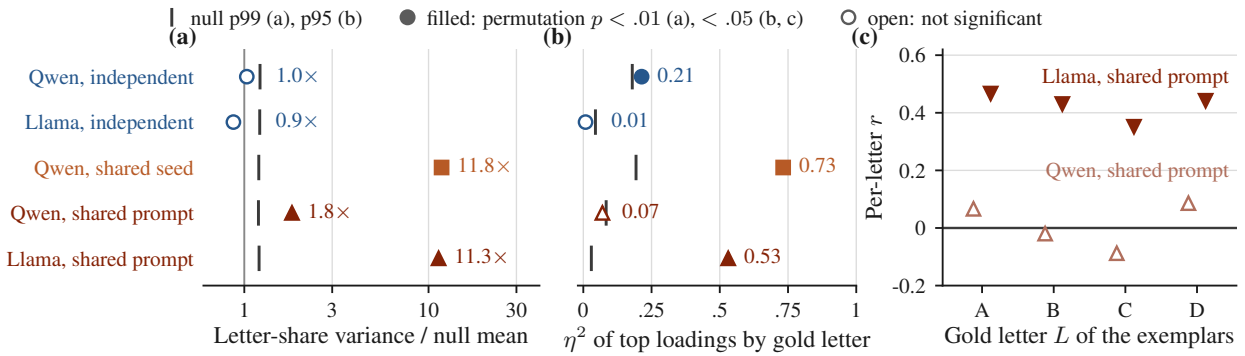


Figure A2. What the shared component couples, for the three shared-component runs and the independent-draw runs of the same models. **(a)** Across-pass variance of a pass's share of each answer letter, summed over the four letters and divided by its mean when each question's answers are shuffled across passes; tick: 99th percentile under shuffling. **(b)** η^2 of the top eigenvector's question loadings by gold letter; tick: 95th percentile with the gold letters permuted. **(c)** Correlation, over passes, between the number of a pass's five exemplars whose answer is letter L and the pass's share of answers L. Filled markers: permutation p below the threshold in the legend.

What the directions follow. With a shared seed, the across-pass variance of the fraction of a pass's answers that are each letter, summed over letters, is 11.8 times its mean when each question's answers are shuffled across passes (1.03 for the independent run), and the top eigenvector's loadings separate the questions by gold letter ($\eta^2 = 0.73$, $p = 0.0002$; Fig. A2a,b). A pass whose noise favors one letter gains on questions with that gold letter and loses on the others, so the coupling cancels in the number correct per pass (observed over predicted variance 0.83). Llama-3-8B with a shared prompt copies the exemplars' letters (Fig. A2c), as few-shot prompts are known to pull answers toward their labels (Zhao et al., 2021): per-letter r between 0.35 and 0.47, letter shares 11.3 times their shuffle mean, $\eta^2 = 0.53$, and an overdispersed number correct per pass (ratio 1.51, $p = 0.001$). Neither direction groups by subject beyond chance ($p = 0.23$ and 0.35). Qwen2.5-7B's loadings with a shared prompt do not group by gold letter ($\eta^2 = 0.07$, $p = 0.094$), and its letter shares do not follow the exemplars' letters (per-letter r between -0.09 and 0.09), but the exemplar set matters: two passes that share an exemplar answer the borderline questions more alike than two that share none, by 10.7 null standard deviations when the exemplar sets are permuted across passes (5.9 for Llama-3-8B; permutation $p = 0.0005$ for both). Its loadings group weakly by subject ($p = 0.042$, uncorrected), and a same-subject exemplar raises accuracy on borderline questions by 3.2 points ($p = 0.033$); only these two effects link the coupling to the exemplars' content. The Qwen2.5-7B independent run shows $\eta^2 = 0.21$ ($p = 0.023$) although its top eigenvalue lies below its edge; we do not interpret it. The answer-letter test of App. C finds no pair of its borderline questions with letters associated beyond chance, but that test looks for single coupled pairs, not for a grouping of the loadings by gold letter.

F. The 24-model ensemble

This appendix lists the ensemble, defines the one-ability nulls and their validity, and gives the analyses behind Sections 3.2 and 3.3.

Roster. Five Qwen (Qwen2.5 at 0.5B, 1.5B, 3B and 7B, and Qwen2-7B), five Mistral (Mistral-7B v0.2 and v0.3, Zephyr-7B-beta, Ministral-8B, Mistral-Small-22B), four Llama (Llama-3.2 at 1B and 3B, Llama-3-8B, Llama-3.1-8B), three Phi (Phi-2, Phi-3-mini, Phi-3.5-mini), two Gemma-2 (2B and 9B), and one each of SmolLM2-1.7B, OLMo-2-7B, Yi-1.5-6B, DeepSeek-LLM-7B and Granite-3.1-8B: ten families, five with a single model. Phi-2 is a base model; the other 23 are instruction-tuned. Greedy accuracy in rotation 0 runs from 35.0% (Llama-3.2-1B) to 72.8% (Gemma-2-9B).

One-ability nulls. Every null is simulated conditional on every kept column staying split, so each null matrix has the observed number of columns and the observed λ_+; each has 1000 draws. Rasch is fitted by joint maximum likelihood. The fixed-margin null draws uniformly among 0/1 matrices with the observed row and column sums. The 2PL treats the abilities as fixed and integrates the question parameters over a normal population (marginal maximum likelihood); it is fitted to all 500 columns and simulated on the split ones. The Bock–Aitkin 2PL integrates the abilities instead, with slopes bounded by 4. The 3PL uses the floor 0.25 for every score. The letter-aware null is a know-or-guess model fitted to rotation 0, with each model's letter preference π_m estimated from all four rotations. The 4-rotation letter know-

or-guess (KG-l) and uniform know-or-guess (KG-u) nulls are fitted to all four rotations jointly, with guessing by π_m or uniform, and simulate every rotation.

Validity. A null whose λ_1 misses the observed one ties correctness to ability more or less tightly than the data, and its higher eigenvalues are then not comparable with the observed ones: the 2PL, Rasch and fixed-margin nulls put the observed rotation-0 λ_1 above all their draws, and the 4-rotation know-or-guess and Bock–Aitkin nulls put the all-4-correct λ_1 below all theirs. We therefore use a null as evidence only if the observed λ_1 lies between its 1st and 99th percentiles, in both directions, for every score and every null, the soft-preference null included. Under this rule the rotation-0 rank 2 is not evidence against the 2PL, and the all-4-correct verdict rests on the 2PL and Rasch alone. The rule was set after the first analyses, and it is necessary, not sufficient (see Size below).

Construction family. All-4-correct is one of several ways to combine a model's four rotations, and a letter preference that acts on choices the model partly knows can leak into any of them. Table A2 gives each score under the nulls valid for it. Rotation 0 has one direction beyond ability, the letter direction of Section 3.2. Without Llama-3.2-1B the per-rank rule finds none under the valid 3PL and letter-aware nulls, but the family-wise test rejects both ($p_T = 0.0035$ and 0.015, 2000 new draws, post hoc); the maximum is at rank 4, ranks 3 and 4 lie at the 98.5th to 99.9th percentiles, and the letter grouping moves to eigenvector 3. Of the rotation-combined scores, all-4-correct has none by the per-rank rule and a marginal family-wise excess (Section 3.3). On ≥ 2 of 4 the per-rank rule finds three under the 2PL (one without Llama-3.2-1B) and none under the 3PL, and the family-wise test agrees ($p_T = 0.002$ and 0.86). On ≥ 3 of 4 the per-rank rule finds two under the Rasch and KG-l nulls and none under the 2PL and KG-u, and none without Llama-3.2-1B; the family-wise test rejects the Rasch, KG-l and 2PL nulls ($p_T = 0.008$, 0.007 and 0.009) but not KG-u (0.28). In the 2000 draws of the family-wise test the per-rank counts on ≥ 3 of 4 differ (Rasch 1, KG-l 3, 2PL 0), a sign of how close these ranks lie to the 99th percentiles. The SP null cannot tell whether letter preference explains the ≥ 3 result, because it is invalid there (Family-wise test below). On ≥ 1 of 4 only the SP null is valid, and it finds nothing (calibrated p 0.750, family-wise $p_T = 0.996$); the mean has no valid null. Removing the 23 questions whose options refer to other options ("all of the above", "A and B only") leaves the counts of rotation 0 and all-4-correct unchanged, lowers those of ≥ 2 (2PL 2) and ≥ 3 (Rasch 1, KG-l 0), and makes the BA null valid on ≥ 1 of 4, with no direction.

Soft-preference null. In this null, model m chooses option j of question i in a rotation with probability proportional to $\exp\big(b_{ij} + \mathbb{1}[j = g_i](a_i\theta_m + e_{mi})\big)\,\pi_m(L_j)$, where g_i is the gold option, $b_{ig_i} = 0$, L_j is the letter at which the rotation shows option j, and $e_{mi} \sim N(0, \tau^2)$ is shared by the four rotations of a cell and independent across models. Ability θ_m is the only link between questions, and the letter preference acts on every choice, also when the model partly knows the answer. The fit is penalized maximum likelihood over the parsed choices of all four rotations ($\tau = 1.17$). Simulations keep the observed unparseable cells, and the rotation-0 result depends on this: without the mask the observed λ_2 lies at the 99.8th percentile of the null, with it at the 95.1st. Refitting inflates the slopes, so we calibrate: 200 datasets simulated from the fit, each refitted with its own null. The calibrated p of rank j is the fraction of these datasets whose λ_j percentile is at least the observed one. Each refitted null uses 1000 draws. Validity follows the rule above. On rotation 0 the SP null is valid (λ_1 at the 85.7th percentile) and rank 2 has calibrated $p = 0.010$ (0.015 by the Monte Carlo convention $(1 + k)/(1 + n)$, with k of the $n = 200$ datasets at least as extreme), but the family-wise p_T is 0.079; without Llama-3.2-1B the null is invalid. On ≥ 3 of 4 and all-4-correct it is invalid (λ_1 at the 0th percentile), so its rejections there are not evidence. On 200 datasets simulated from the fit, each with the null refitted and 200 draws, the valid rotation-0 nulls reject rank 2 in 60% (letter-aware) to 99.5% (3PL) of datasets. The 2PL rejects rank 2 of ≥ 2 of 4 in 67.5% of datasets at the fitted parameters, 39% after a bootstrap correction of the parameters' bias and 72% after rescaling their mean squares; the intervals cover Monte Carlo error only, and why the bias correction lowers the rate is not identified.

Predictive check. The soft-preference model fails its preset predictive check, which compares 500 datasets simulated from the fit with the data. Per-model accuracy and letter shares fit, but the number of rotations, 0 to 4, in which a model answers a question correctly is covered for 82.5% of model-by-count cells, below the required 90%. The weakest, letter-biased models have too many questions correct in exactly one rotation, as expected if they choose one letter almost deterministically, and stronger models too many correct in all four (pooled: 0.196 observed against 0.177 simulated for one rotation, 0.382 against 0.366 for four). The calibration corrects the bias of the fitting pipeline, not this misfit, so the soft-preference verdicts are not fully validated.

Family-wise test. The per-rank rule tests each rank at 1% and stops at the first rank that passes, so it can miss an excess spread over several ranks. An audit of an earlier draft of this analysis had raised this and had computed a similar statistic, the largest percentile over ranks 2–4, on the same data, and its model-pair correlations had pointed to the sibling models. We then wrote down the family-wise test, the sibling-model

Table A2. Construction family, soft-preference null and power. Each score combines a model's four rotations: correct in rotation 0, in at least k of 4, or the mean. Valid nulls: those whose 1st–99th percentile range contains the observed λ_1, each with its number of directions beyond ability by the per-rank rule ("none": no valid null other than SP), for all 24 models and without Llama-3.2-1B (w/o); LA: letter-aware; KG-l, KG-u: 4-rotation letter and uniform know-or-guess; BA: Bock–Aitkin 2PL. Soft-preference (SP) null: rank-2 calibrated p; "inv.": the null fails the validity rule. Power: detection rate of a planted second ability at $s = 2$ by the SP null (valid, and rank 2 above its calibrated threshold) and by the 2PL per-rank rule, 150 datasets per s; –: not simulated or not valid. [b]False-positive rate 0.69 without a second factor. [c]From the 200 datasets of Table A4. The per-rank counts use 1000 draws per null; the family-wise p_T of the valid nulls (2000 new draws) are given in the text.

Score	Cols	λ_1–λ_4	Valid nulls: ranks beyond ability 24 models	w/o Llama-1B	SP: rank-2 cal. p 24 / w/o	Power, $s = 2$ SP / 2PL
Rotation 0	460	85.9, 49.9, 29.7, 27.3	3PL 1, KG-u 1, LA 1	3PL, LA 0	0.010 / inv.	0.64 / –
≥ 1 of 4	334	34.2, 27.2, 23.7, 21.4	none	BA 0	0.750 / inv.	–
≥ 2 of 4	444	108.3, 42.3, 30.2, 23.0	2PL 3, 3PL 0	2PL 1, 3PL 0	0.245 / 0.280	0.85 / 0.87[b]
≥ 3 of 4	439	103.4, 37.8, 25.5, 22.1	Rasch 2, KG-l 2; 2PL, KG-u 0	2PL, KG-l, KG-u 0	inv. / 0.285	–
All 4	391	92.1, 29.0, 23.3, 20.8	Rasch, 2PL 0	Rasch, 2PL, KG-u 0	inv. / inv.	0.45 / 0.81[c]
Mean	489	127.5, 43.1, 28.5, 25.7	none	none	inv. / inv.	0.19 / –

test and the single validity rule before running them, with seeds disjoint from all earlier runs. The specification fixes the statistic, the ranks, the decision rule and the groups; it does not make the hypothesis independent of the data, which had suggested it. The statistic T is the largest of the percentiles of λ_2, λ_3 and λ_4 among 2000 new null draws (ties count half); each draw's own T_d is computed against the other draws, and $p_T = (1 + \#\{d : T_d \geq T\})/2001$, at level 0.05. Table A4 gives the results. On 200 datasets simulated from the SP fit with no second factor (500 null draws each), the test never rejects all-4-correct under the 2PL or Rasch; the SP model is invalid on all-4-correct, so this says nothing about the size of the test, which the next paragraph measures. On ≥ 2 and ≥ 3 of 4, the refitted 2PL null is valid in 98.5% and 100% of the datasets, and it is valid and rejects in 98.5% and 58% of all datasets. The refitted Rasch null is valid in 0% and 18% of the datasets; it is valid and rejects in 0% and 11% of all datasets, which on ≥ 3 is 61% of the datasets in which it is valid. The SP fit is valid on ≥ 2 but not on ≥ 3, so the ≥ 3 rates are indicative only, and where $T = 100$ the p lies on a coarse grid. Of the observed ≥ 3 rejections (p_T 0.007 to 0.009, Construction family above), the 2PL one occurs in 58% and the Rasch one in 11% of datasets from a fit that the validity rule rejects on ≥ 3, and no simulation covers the KG-l test; we draw no conclusion from ≥ 3 of 4. In the new 2PL draws $\lambda_4 = 20.83$ is above the rank-4 99th percentile (20.80), as in replicate B, against 20.85 in the 1000 draws of Fig. 1d (Monte Carlo $p = 0.012$).

Size. The following checks were specified after the family-wise result and are post hoc. We simulate 300 all-4-correct datasets from each fitted one-ability model, the 2PL and Rasch, with the nulls' own simulator, refit both nulls to each dataset (500 draws each), and apply the validity rule and the family-wise test (Table A4). The 2PL test rejects in 6.7% of datasets under 2PL truth and 8.7% under Rasch truth (95% intervals 4.1–10.1% and 5.7–12.4%), so it is not size-controlled by the bound of 7.5% we set before the run. The Rasch test rejects in 6.7% and 0.7%, but under Rasch truth mainly through the validity screen: the refitted Rasch null is valid in only 23% of those datasets. Ranking the observed p_T among the simulated ones gives calibrated values of 0.033 (2PL truth) and 0.047 (Rasch truth) for the 2PL test and 0.057 and 0.017 for the Rasch test. The validity rule is therefore necessary, not sufficient: it does not detect wrong slopes (the Rasch null is valid in every dataset simulated from the 2PL fit), and it rejects the correct Rasch model, refitted by joint maximum likelihood, in most datasets simulated from it.

Sibling models. Two models are siblings if they are versions or fine-tunes of one base model at one parameter count, decided from model metadata before the run: Phi-3-mini and Phi-3.5-mini; Meta-Llama-3-8B and Llama-3.1-8B; Mistral-7B v0.2, v0.3 and Zephyr-7B-beta; Qwen2-7B and Qwen2.5-7B. Models of one family at different sizes are not siblings. Each of the $2 \cdot 2 \cdot 3 \cdot 2 = 24$ subsets that keep one model per group has 19 models; both nulls are refitted to each, with 1000 draws. The 24 subsets give p_T 0.09 to 0.69 and none reject. The control removes five of the 15 models outside the groups, chosen at random, 24 times, so that 19 models remain as well; all 24 reject ($p_T \leq 0.047$). The subsets of each kind overlap heavily, so the comparison is one contrast, not 24 replicates. The random removals include models that share a developer with the groups (Qwen2.5 at 0.5B to 3B, Llama-3.2, Ministral-8B and Mistral-Small), and they are not matched to the 7B–8B chat models of the groups. On all-4-correct both nulls are valid in every sibling and random subset. In rotation 0 the 2PL and Rasch nulls are valid in none of the 24 sibling subsets and the uniform know-or-guess null in 17; the 3PL and letter-aware nulls

are valid and reject in all 24, so the letter direction does not depend on the siblings.

Pairs and simple baselines. The sibling result does not say which groups matter. In a post hoc follow-up we fixed two pairs, Phi-3 and Phi-3.5-mini and Meta-Llama-3-8B and Llama-3.1-8B, and compared the four subsets that remove one model of each pair with all 190 subsets that remove two of the other 20 models (1000 draws per subset; every null valid in every subset). All four pair removals lie above every reference removal under both nulls (Table A4); 90% (2PL) and 74% (Rasch) of the reference removals still reject. Removing Qwen2-7B or Qwen2.5-7B makes the rejection stronger (mean reference p_T 0.010 and 0.006 under the 2PL), so the Qwen pair is not part of the excess. The effect is specific to the two pairs: removing the Mistral-7B v0.2 and v0.3 near-duplicates (residual rank 3 in Table A3) keeps the rejection ($p_T = 0.019$ under both nulls), as do the other two removals within the Mistral group (0.004 to 0.023). Nor does removing two strong chat models explain it: the 15 reference removals whose two models have rotation-0 accuracy between 0.62 and 0.71 (the pair models: 0.644 to 0.684; no two other models lie inside that narrower range) give p_T at most 0.052 (2PL) and 0.113 (Rasch), below every pair removal. The letter-biased model contributes too. Without Llama-3.2-1B alone (post hoc, 2000 new draws), p_T is 0.06 (2PL) and 0.089 (Rasch), and 0.15 under the uniform know-or-guess null, which becomes valid; the secondary statistic of the specification, $\lambda_2 + \lambda_3 + \lambda_4$ against its null draws, on which no decision rests, still gives p 0.023 and 0.016. Of the 19 reference removals that include Llama-3.2-1B, 12 (2PL) and 6 (Rasch) still reject, with p_T at most 0.099 and 0.130. Removing Llama-3.2-1B together with one model of each pair gives 0.30 to 0.59 and 0.38 to 0.60 (1000 draws), close to the pair removals alone. Table A3 gives two simple baselines. Letter shares in rotation 0 flag the three models whose most frequent letter exceeds half of their replies; these shares count unparseable replies in the denominator, whereas the preference of 0.90 for A in Section 3.2 is estimated over guesses. The correlations of model pairs' residuals (observed minus 2PL-expected correctness over the 391 all-4-correct columns) rank the Phi and Llama pairs first and second of 276. These tables find the same candidates as the spectrum; the spectrum adds a test of whether they exceed one ability, and the rotations attribute the larger structure to letter preference.

Power. To measure what the all-4-correct verdict can exclude, we add a second ability $\theta_{2,m} \sim N(0, 1)$, independent of θ_m, with slope s to the gold-option term $a_i\theta_m + e_{mi}$ of the 180 STEM questions, simulate datasets from the soft-preference fit, and refit every null. The fit's median slope $|a_i|$ is 1.08, so $s = 1$ is about a typical first-ability slope; at $s = 2$ the planted term carries on average 60% of the variance across models of a STEM question's gold-option term (30% at $s = 1$). Table A4 gives the all-4-correct power from 200 datasets per s. The datasets come from the soft-preference fit, which is invalid on all-4-correct (λ_1 below all its draws), so these rates are indicative, not a measurement for data like ours. An earlier simulation with 150 datasets per s gives the power column of Table A2; in it the 2PL per-rank rule never fires on all-4-correct without a planted factor, but fires in 69% of ≥ 2 datasets. The SP rule fires in at most 2% of datasets at $s = 0$.

Table A3. Simple baselines. Top: rotation-0 letter shares (unparseable replies in the denominator) of the models whose largest share exceeds 0.5; every other model has a largest share of at most 0.35. Bottom: correlation of model pairs' residuals after the all-4-correct 2PL fit, with rank among 276 pairs (mean -0.04, standard deviation 0.08).

Model	A	B	C	D
Llama-3.2-1B	0.81	0.04	0.09	0.05
Qwen2.5-0.5B	0.23	0.06	0.12	0.58
SmolLM2-1.7B	0.07	0.20	0.17	0.56

Model pair	r	**Rank**
Phi-3-mini, Phi-3.5-mini	0.40	1
Meta-Llama-3-8B, Llama-3.1-8B	0.27	2
Mistral-7B v0.2, v0.3	0.20	3
Mistral-7B v0.3, Meta-Llama-3-8B	0.19	4
Mistral-7B v0.2, Zephyr-7B	0.17	5
Qwen2-7B, Qwen2.5-7B	0.03	37

Letter preference. Figure A3 shows the evidence of Section 3.2. The original-option correlation in panel (b) follows from the shown-letter correlation and the cyclic shape of the letter profile, so it adds no separate evidence; the same-question correlation does. Eigenvector 1 also carries a letter part ($\eta^2 = 0.22$ by gold letter), and eigenvectors 3 and 4 group only slightly ($\eta^2 = 0.087$ and 0.061). The same models carry eigenvector 2 in every rotation: their scores correlate with those of rotation 0 at $|r| = 0.95$ to 0.98. Without Llama-3.2-1B, refitted on 23 models, no rank from 2 upward exceeds the 3PL or letter-aware 99th percentile except rank 4, by 0.35 and 0.57; the family-wise test rejects both nulls (Construction family above), and the letter grouping moves to eigenvector 3 ($\eta^2 = 0.21$ by gold letter). In expectation, benchmark-level scores cannot carry a letter preference when gold letters are balanced, as nearly here (118, 124, 129, 129 for A to D).

The remaining eigenvectors. The all-4-correct eigenvector 2 groups its question loadings by subject ($\eta^2 = 0.21$, $p = 0.001$), but a cubic in question difficulty explains 39% of its loading variance, and after removing it the grouping is at chance ($p = 0.24$). Of the borderline directions of the other scores, only ≥ 2 ranks 2 and 3 group by subject

Table A4. Family-wise test on all-4-correct (2PL and Rasch nulls; reject at $p_T < 0.05$). Sibling and random subsets: subsets that reject, and the range or maximum of p_T. Pairs: the four removals of one Phi and one Llama model, and the 190 removals of two other models (fraction that reject, maximum p_T). Unparseable: replies imputed, the 95 questions whose score they affect dropped, or all 217 questions with any dropped (1000 draws each). Size: rejection rate in 300 datasets simulated from each fitted one-ability model, and the observed p_T calibrated against them (2PL truth, Rasch truth). Power: family-wise / per-rank rule, 200 datasets per slope s.

	2PL	**Rasch**
Pct. of $\lambda_2, \lambda_3, \lambda_4$	71.8, 98.5, 99.1	94.6, 97.0, 98.6
p_T, 24 models	0.026	0.041
Sibling subsets	0/24 (0.09–0.65)	0/24 (0.12–0.69)
Random subsets	24/24 ($\leq$0.047)	24/24 ($\leq$0.046)
Pair removals	0/4 (0.22–0.69)	0/4 (0.32–0.38)
Other two removed	0.90 ($\leq$0.099)	0.74 ($\leq$0.133)
Unparseable: impute	0.019	0.042
drop 95	0.036	0.043
drop 217	0.23	0.12
Size, 2PL truth	0.067	0.067
Size, Rasch truth	0.087	0.007
Calibrated p_T	0.033, 0.047	0.057, 0.017
Power, $s = 1$	0.34 / 0.06	0.21 / 0.08
Power, $s = 2$	0.92 / 0.81	0.87 / 0.80

after this removal ($p = 0.010$ and 0.034, uncorrected over eight eigenvectors). The models' scores on the all-4-correct eigenvector 4, below λ_+, group by family ($\eta^2 = 0.74$, $p = 0.006$); in rotation 0, after a Holm correction, only eigenvector 4 aligns with family ($p = 0.017$), and same-family model pairs correlate little more than other pairs (mean r 0.414 against 0.375, $p = 0.085$ by a permutation of model labels).

Replicates and near ties. Replicate B differs from A in the chosen option in 6 of 48,000 cells, so it tests GPU nondeterminism, not resampling; the largest eigenvalue difference between the replicates over every matrix compared is 0.081. A near tie is a cell whose two most probable first-token letters differ by less than 0.13 nats in log probability, in either replicate (5.0% of cells). Setting near ties to the majority answer over rotations leaves every verdict unchanged. Dropping the 297 questions with a rotation-0 near tie leaves 203, with less power; rank 2 then falls below the letter-aware 99th percentile but stays above the 3PL's. Both nulls are valid for this variant (observed λ_1 near their 13th and 14th percentiles).

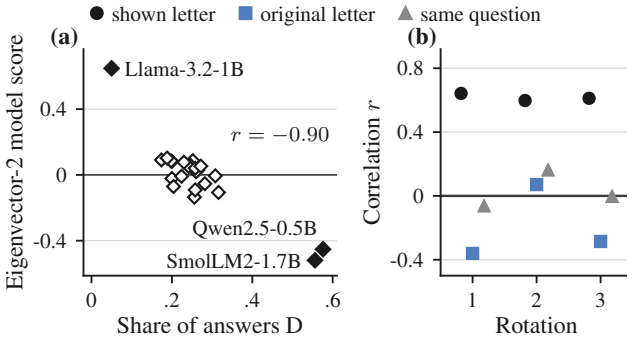


Figure A3. Eigenvector 2 of the 24-model matrix is answer-letter preference. **(a)** Each model's score on eigenvector 2 in rotation 0 against its fraction of answers that are D ($r = -0.90$, permutation $p = 10^{-4}$); filled: the three models at the ends. **(b)** For rotations 1–3, correlation of each question's loading on eigenvector 2 with the rotation-0 mean loading of a gold letter, taken at the letter now shown (shown letter) or at the original letter (original letter), and with the question's own rotation-0 loading (same question); 440, 433 and 448 questions.